ECE343 Spring 2025

Objective 7.4: Magnetic Field Boundary Conditions



UNITED STATES
AIR FORCE
ACADEMY

Lesson	Topics	Reading	Notes
24	BVP (Capacitor, Coax), Electric Potential	5.16	LO 5.3
25	Electric Boundary Conditions	5.18-5.22	LO 5.4
26	BC's/Capacitance	5.23-5.24	
27	Electrostatic Lab	Lab packet	
Chapter 6: Steady Current & Conductivity			
28	Current density, Conductivity, Resistance	Ch. 6	
Chapter 7: Magnetostatics			
29	Magnetic Fields, Gauss' Law for Magnetism	7.1-7.3	LO 7.1-7.2 **GR 2**
30	Ampere's Law	7.4-7.5, 7.9	LO 7.3
31	Ampere's Law Applied	7.6-7.8	LO 7.3
32	Magnetic Boundary Conditions	7.10-7.14	LO 7.4
33	Magnetostatics Lab	Lab packet	
Chapter 8: Time-Varying Fields			
34	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.1
35	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.2-8.3
Chapter 9: Plane Waves in Lossless Media			
36	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
37	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
38	Uniform Plane Waves & Polarization	9.4-9.6	LO 9.2-9.3
39	Normal Incidence	Vol 2, 5.6-5.7	LO 9.4-9.5
40	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
41	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
42	Incidence Lab	TBD	LO 9.5

Ch 7 Learning Objectives

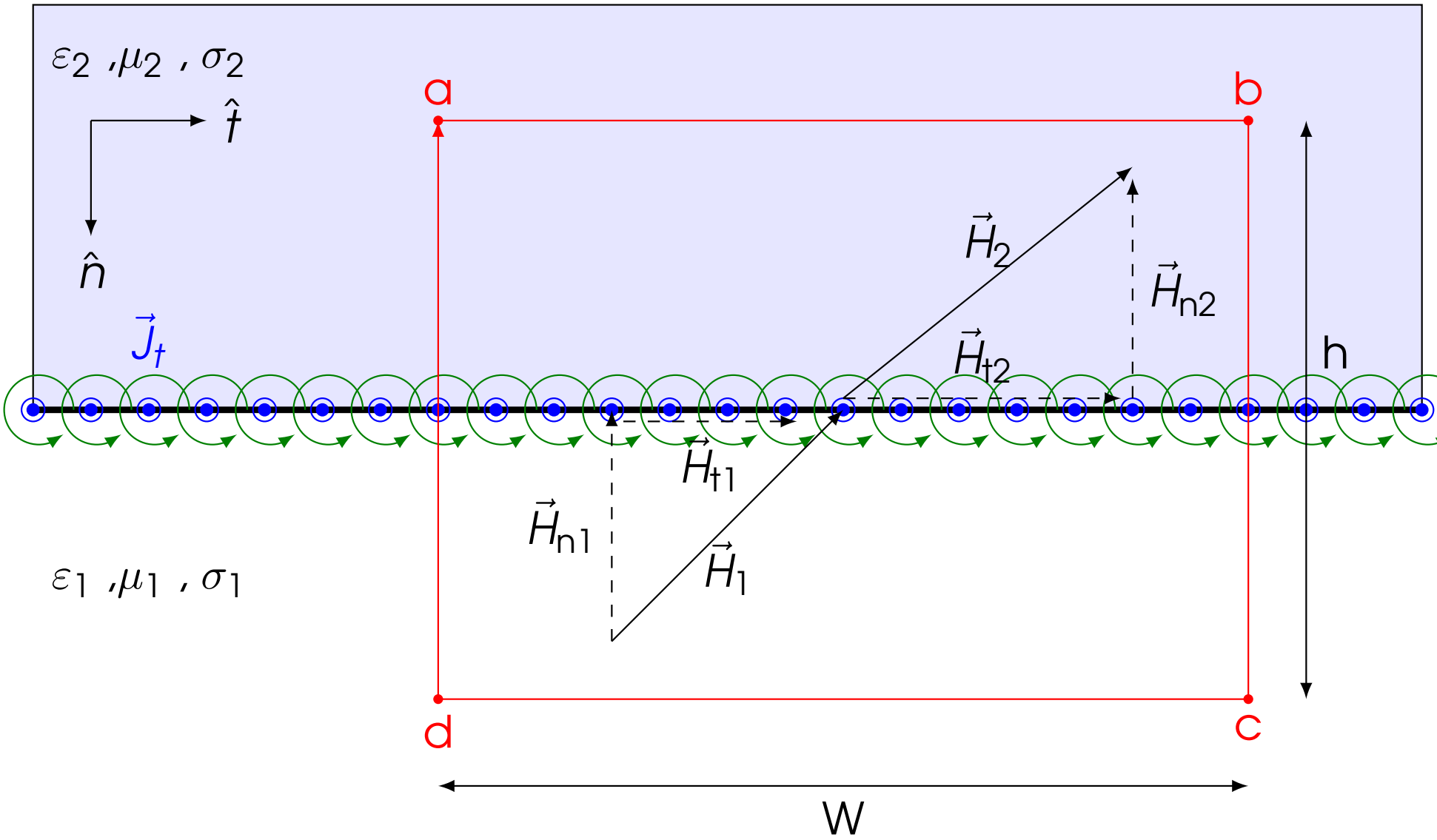
7.1 I can use the Biot-Savart law to simple magnetostatic systems to find the magnetic field at an observation point.

7.2 I can articulate the meaning of both the integral form and point form of Gauss' Law for magnetism.

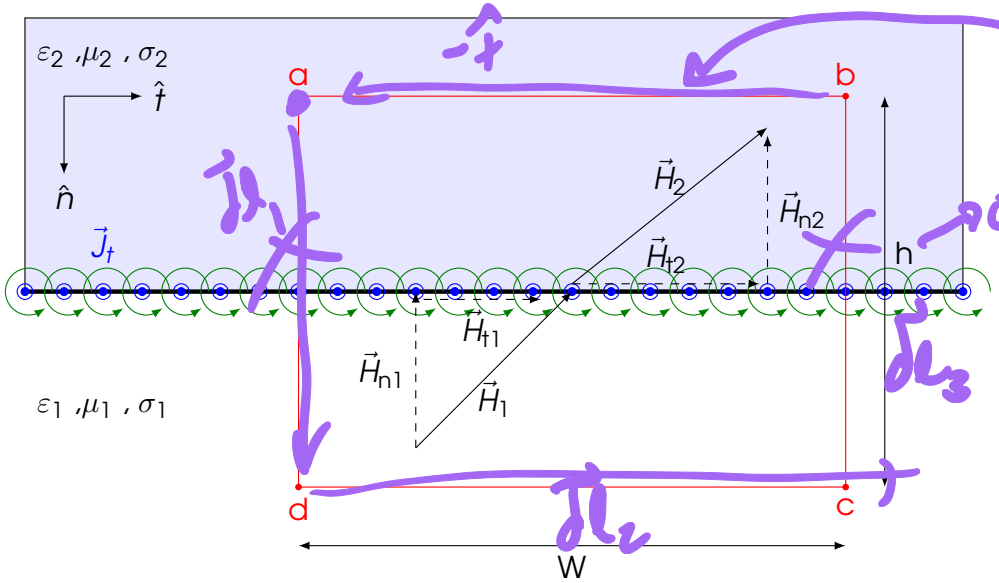
7.3 I can use Ampere's Law to calculate the magnetic field at an observation point for simple magnetostatic systems.

7.4 I can apply the magnetic field boundary conditions to calculate magnetic fields on both sides of a material boundary.

Magnetic Boundary Conditions



$\vec{H}$ Boundary Conditions: Transverse



We can write:

① Draw a picture, incl. amp. path

② $\oint_C \vec{H} \cdot d\vec{\ell}$

$= \int_c^d \vec{H} \cdot d\vec{\ell}_1 + \int_d^c \vec{H} \cdot d\vec{\ell}_2$

$+ \int_b^a \vec{H} \cdot d\vec{\ell}_3 + \int_a^b \vec{H} \cdot d\vec{\ell}_4$

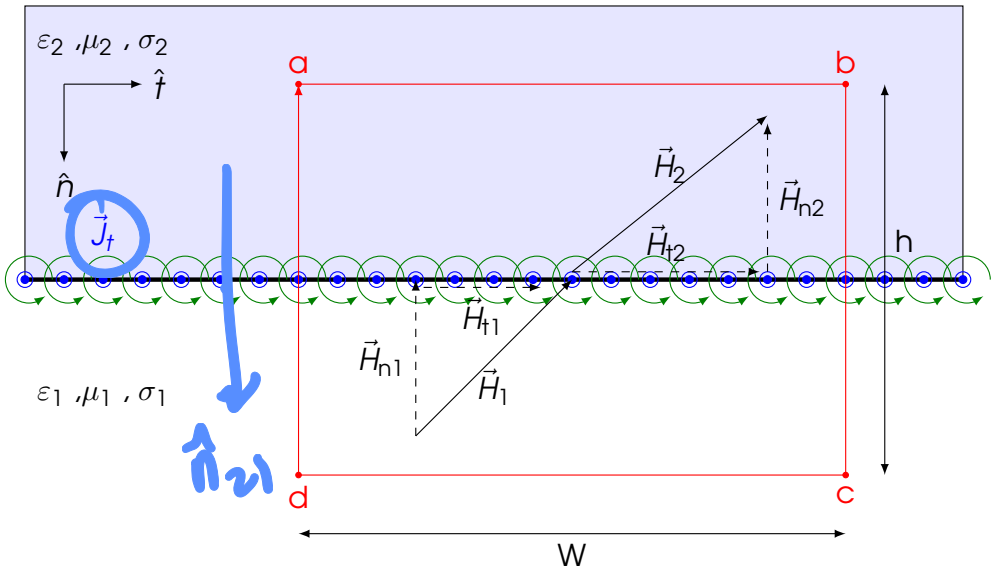
$= \int_c^d \vec{H}_1 \cdot d\vec{\ell}_2 + \int_b^a \vec{H}_2 \cdot d\vec{\ell}_4$

$= \vec{H}_{t1} W - H_{t2} W$

Using Ampere's Law for magnetism:

$\oint_C \vec{H} \cdot d\vec{\ell} = I_{encl.}$

$\vec{H}$ Boundary Conditions: Transverse



We can write:

③ $I_{encl.} = \vec{J}_t W$ $\frac{A}{m} \cdot m = A$

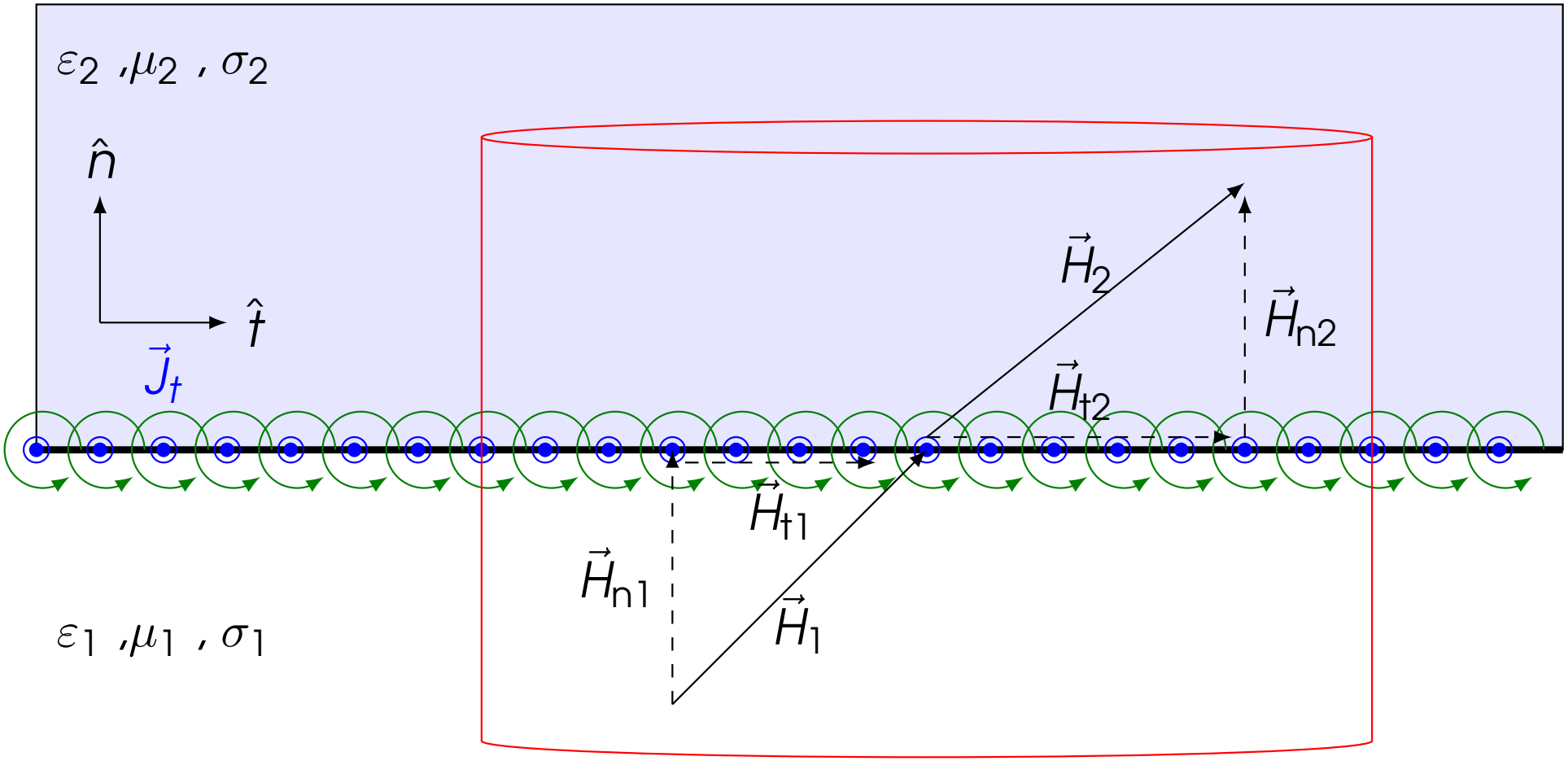
④ $\vec{H}_{1t} - \vec{H}_{2t} = \vec{J}_t$

$\hat{n}_{21} \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_t$

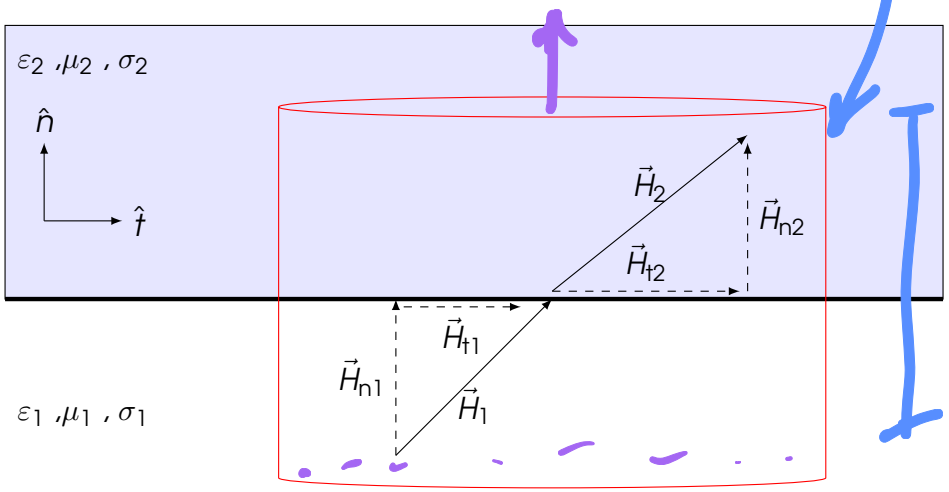
Using Ampere's Law for magnetism:

$\oint \vec{H}_t \cdot d\vec{\ell} = I_{encl.}$

$\vec{H}$ Boundary Conditions: Normal



$\vec{H}$ Boundary Conditions: Normal



Gaussian Surface $= \iint_{Top} \vec{B}\cdot d\vec{s} + \iint_{Bottom} \vec{B}\cdot d\vec{s}$

$+ \iint_{Sides} \vec{B}\cdot d\vec{s}$ $h \to 0$

$h \to 0$

$= B_{n2} A_{Top} - B_{n1} A_{Bott}$

$= 0$

Gauss' Law:

$\oint \vec{B}\cdot d\vec{s} = 0$

$B_{n2} = B_{n1}$

$$\mu_2 H_{n2} = \mu_1 H_{n1}$$

$\vec{H}$ Boundary Condition Summary

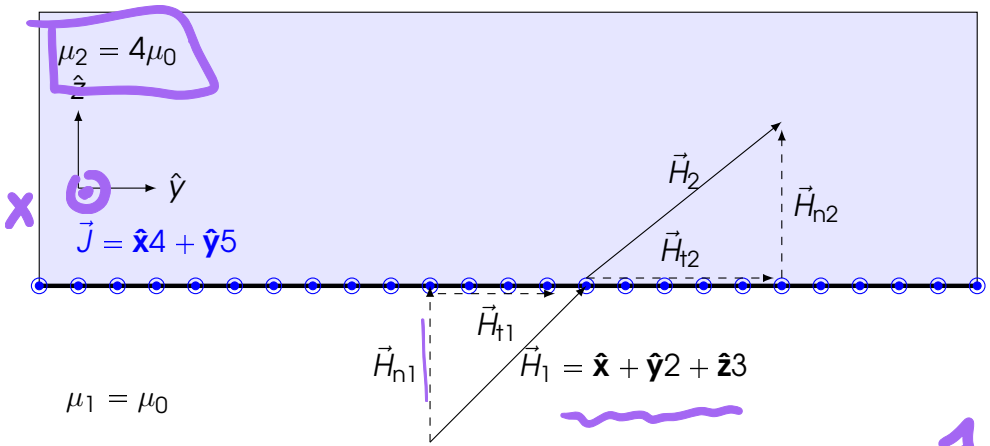
	Electric	Magnetic
Normal	$\hat{n}_{21} \cdot \left(\varepsilon_1 \vec{E}_1 - \varepsilon_2 \vec{E}_2\right) = \rho_s$	$\mu_2 \vec{H}_{n2} = \mu_1 \vec{H}_{n1}$
Transverse	$\vec{E}_{t1} = \vec{E}_{t2}$	$\hat{n}_{21} \times \left(\vec{H}_1 - \vec{H}_2\right) = \vec{J}$

$\hat{n}_{21} \cdot (\vec{H}_1 - \vec{H}_2) = J_{s1}$

$\hat{n}_{21} \times (\vec{E}_1 - \vec{E}_2) = 0$

Boundary Condition: Example

A magnetic field $\vec{H}_1 = \hat{\mathbf{x}} + \hat{\mathbf{y}}2 + \hat{\mathbf{z}}3$ exists in a material with $\mu = \mu_0$ and is incident on a boundary with a surface current of $\vec{J} = \hat{\mathbf{x}}4 + \hat{\mathbf{y}}5$. The material on the other side of the boundary has $\mu = 4\mu_0$. Determine $\vec{H}_2$.



1. Draw a picture
2. Identify T & N
   $\vec{H}_{t1} =$
   $\vec{H}_{n1} =$
3. $\hat{n}_{2,1} =$
4. Normal BC

Transverse

$$\hat{z} \times (\vec{H}_1 - \vec{H}_2) = \vec{J}$$

$$-\hat{z} \times (\hat{x} + \hat{y}2 + \hat{z}3 - \hat{x}H_x + \hat{y}H_y - \hat{z}H_z)$$

$$-\hat{y} + \hat{x}2 + \hat{y}H_x - \hat{x}H_y = \hat{x}4 + \hat{y}5$$

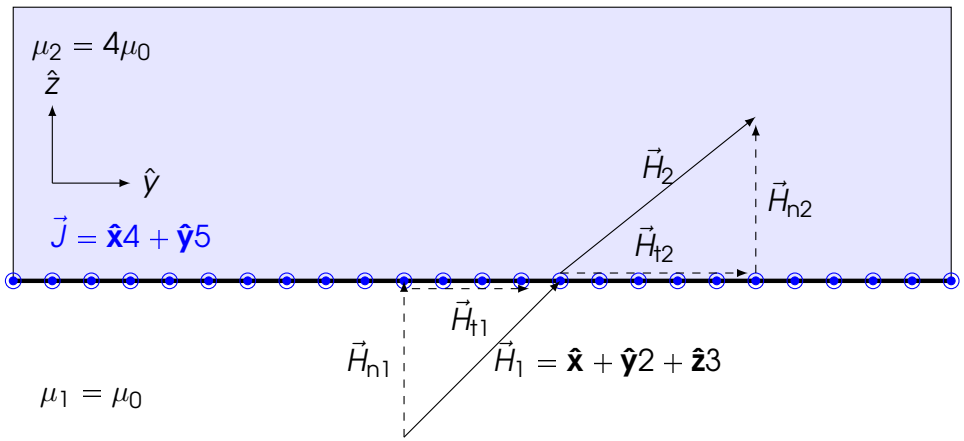
$$\hat{x} \rightarrow 2 - H_y = 4$$

$$\Longrightarrow H_{2y} = -2$$

$$\hat{y} \rightarrow -1 + H_x = 5 \Longrightarrow H_{2x} = 6$$

Boundary Condition: Example

A magnetic field $\vec{H} = \hat{\mathbf{x}} + \hat{\mathbf{y}}2 + \hat{\mathbf{z}}3$ exists in a material with $\mu = \mu_0$ and is incident on a boundary with a surface current of $\vec{J} = \hat{\mathbf{x}}4 + \hat{\mathbf{y}}5$. The material on the other side of the boundary has $\mu = 4\mu_0$. Determine $\vec{H}_2$.



5. Normal BC

$M_1 H_{1z} = M_2 H_{2z}$

$M_0 3 = 4 M_0 H_{2z}$

6. Put it all together

$\vec{H}_2 = \hat{x}6 - \hat{y}2 + \hat{z}\frac{3}{4}$